Caleb Kusumo

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EDUCATION

Boston University, College of Engineering — Boston, MA
B.S. Mechanical Engineering; Minor: Electrical Engineering — Expected May 2029
- GPA: 3.81 | Dean's List, Fall 2025

ENGINEERING EXPERIENCE

Terrier Motorsport Formula SAE | Steering Lead & Aero Team Member — Sep 2025 - Present
- Lead 3+ designers developing TM02 steering: custom rack-and-pinion assembly and right-angle gearbox integration.
- Iterate Ackermann geometry and verify chassis fit in Fusion 360; size 4130 steering tubing for 100 N·m torque.
- Size tie-rod tubing using analytical buckling and SolidWorks Simulation; factor of safety 1.47 at 2,000 N.
- Fabricate fiberglass aerodynamic components and integrate bodywork into the team's formula-style electric race car.
- Contribute to the team's seventh-place finish in the Formula Hybrid + Electric EV class in 2026.

FIRST Robotics Team 6560 — Charging Champions | Captain & Manufacturing Lead — Sep 2022 - Apr 2025
- Led fabrication group of 15+ students in milling, turning, CNC machining, and laser cutting.
- Coordinated design, fabrication, and assembly across a 30+ member team to deliver a robot in under four weeks.
- Served on the drive team; earned Arizona North second place, OC Regional finalist honors, and Innovation in Control, Autonomous, and Creativity awards.

PROJECTS & ENTREPRENEURSHIP

CalebColor Film | Founder & Lead Technician — Sep 2023 - Present
- Founded and operate a film sales and processing business; developed 200+ rolls and sold 400+ rolls within one year.
- Diagnose and repair film/digital cameras: shutter timing, film advance, screens, mainboards, shutters, and light seals.

Lutron Lighting Competition | Third Place — Feb 2026
- Designed, built, wired, and presented a 1:100 modular gallery lighting model; placed third among 40 entries.
- Modeled components in Fusion 360 and fabricated them using 3D printing, waterjet cutting, and laser cutting.
- Integrated custom Hall-effect circuitry to trigger LED strips powered by an onboard battery.

4x5 Monorail Camera | Personal Project — Sep 2026
- Designed and fabricated a 4×5 monorail camera in SolidWorks with a Graflok film-back interface, spring-loaded retention, and dual-purpose bellows clamps to reduce part count.

SKILLS

CAD & Software: SolidWorks (CAD/FEA), Fusion 360 (CAD/CAM), Excel, Adobe Lightroom/Photoshop/Illustrator
Manufacturing: CNC mill/lathe, milling, turning, waterjet, laser cutting, SLA/FDM 3D printing, vacuum forming, composites
Electronics: Soldering, Arduino, Raspberry Pi, oscilloscope operation, sensor integration

LEADERSHIP & SERVICE

Engineering Product Innovation Center Student Advisory Board | Member — Sep 2026 - Present
- Organize machine-shop workshops and community events.

Boston University Photography Club | Darkroom Manager — Sep 2026 - Present
- Manage darkroom facilities and chemical inventory; host public photography, film processing, and printing workshops.

Boston University Technology Innovation Scholars Program | Mentor — Sep 2026 - Present
- Design "Intro to MechE" course for middle school students to build on prior $25K revenue to double outreach capacity.

CERTIFICATIONS

Certified Professional in Design for Manufacturing, Autodesk — Apr 2026